Chapter 12 Vector-Valued Functions (Vector Function)

12.1 Curve in Space and their tangents

In general, a *function* is a rule that assigns to each element of its domain D exactly one element of its range R. A function f is a ***real-valued function*** or ***scalar function*** if its value is a scalar.

Definition 1: Vector-valued function (vector function)
A ***vector-valued function*** or ***vector function*** $\mathbf{r}$ on a domain set D is a rule that assigns a vector in space to each element in D. That is, it is simply a function whose domain is a set of real numbers and whose range is a set of vectors.

For every number t in the domain of $\mathbf{r}$ there is a unique vector in space denoted by $\mathbf{r}(t)$.

Remark: If $f(t)$, $g(t)$ and $h(t)$ are components of the vector $\mathbf{r}(t)$, then f, g and h are scalar functions called the **component functions** (or **components**) of $\mathbf{r}$ and we write

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}\,.$$

The vector function's domain is assumed to be the common domain of the components.

Example 1:
$\mathbf{r}(t) = 2\mathbf{i} + t\ \mathbf{j} + \sqrt{t}\ \mathbf{k}$, $t \geq 0$, is a vector-valued function with domain $[0, \infty)$.

$\mathbf{r}(0) = 2\mathbf{i}$, $\mathbf{r}(1) = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{r}(4) = 2\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$ are some vectors in its range.

Example 2:
If $\mathbf{r}(t) = t^3\mathbf{i} + \ln(3-t)\mathbf{j} + \sqrt{t}\ \mathbf{k}$, then the component functions are

$$f(t) = t^3,\ g(t) = \ln(3-t) \text{ and } h(t) = \sqrt{t}\,.$$

The domain D of r consists of all values of t for which the expression for $r(t)$ is defined.
The functions f, g and h are all defined when $3 - t > 0$ and $t \geq 0$.
$\therefore\ D = [0, 3)$.

There is a close connection between continuous vector functions and space curves.
Suppose that f, g and h are continuous scalar functions on an interval I.
Then the set C of all points (x, y, z) in space, where

$$\boxed{x = f(t),\ y = g(t),\ z = h(t)} \quad \text{---------- (1)}$$

and t varies throughout I, is called a **space curve**.
The equations in (1) are called **parametric equations of *C*** and t is called a **parameter**.
We can think of C (the particle's **path**) as the curve being traced out by a moving particle whose position at time t is the point $P(f(t), g(t), h(t))$ and $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} = \overrightarrow{OP}$ is the particle's **position vector** (see **Figure 1**).

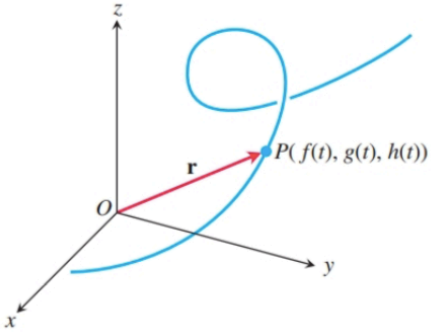


Figure 1 The position vector $\mathbf{r}(t) = \overrightarrow{OP}$ of a particle moving through space is a function qf time.

Figure 2 displays several space curves generated by a computer graphing program.

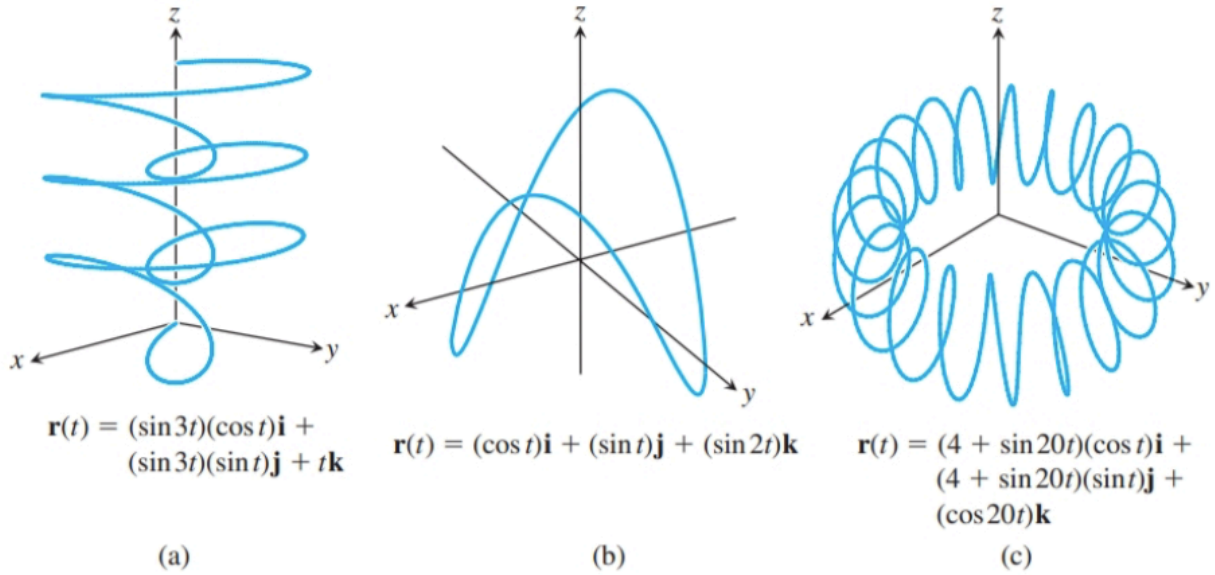


Figure 2 Space curves are defined by the position vectors $\mathbf{r}(t)$.

Example 3:
Graph the vector function $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$.

Solution:
The parametric equations for this curve are

$$x = \cos t, \quad y = \sin t, \quad z = t, \quad t \in \mathrm{R}.$$

Since $x^2 + y^2 = \cos^2 t + \sin^2 t = 1$, the curve must lie on the circular cylinder $x^2 + y^2 = 1$.

Since $z = t$, the curve spirals upward around the cylinder counterclockwise as t increases.

The curve, shown in **Figure 3** is called a **helix**.

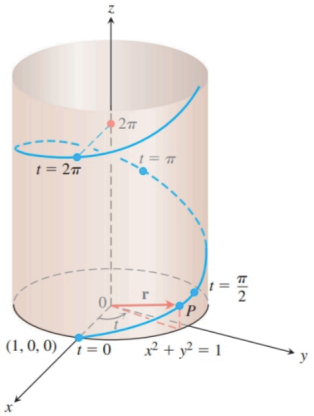


Figure 3 The upper half of the helix $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$

Limits and Continuity of vector function

Definition 2: Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ be a vector function with domain D, and let $\mathbf{L}$ be a vector. We say that $\mathbf{r}$ has **limit L** as t approaches t_0 and write

$$\lim_{t \to t_0} \mathbf{r}(t) = \mathbf{L}$$

if, for every number $\varepsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all $t \in D$

$$|\mathbf{r}(t) - L| < \varepsilon \quad \text{whenever} \quad 0 < |t - t_0| < \delta.$$

To calculate the limit of a vector function, we find the limit of each component scalar function.

$$\lim_{t \to t_0} \mathbf{r}(t) = \lim_{t \to t_0} (f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k})$$

$$= (\lim_{t \to t_0} f(t))\mathbf{i} + (\lim_{t \to t_0} g(t))\mathbf{j} + (\lim_{t \to t_0} h(t))\mathbf{k}$$

Example 4: If $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$, then

$$= (\lim_{t\to\pi/4} \cos t)\mathbf{i} + (\lim_{t\to\pi/4} \sin t)\mathbf{j} + (\lim_{t\to\pi/4} t) = \frac{\sqrt{2}}{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j} + \frac{\pi}{4}\mathbf{k}.$$

We define continuity for vector functions the same way we define continuity for scalar functions defined over an interval.

Definition 3: A vector function $\mathbf{r}(t)$ is continuous at a point $t = t_0$ in its domain if

$$\lim_{t\to t_0} \mathbf{r}(t) = \mathbf{r}(t_0).$$

The function is continuous if it is continuous at every point in its domain.

Example 5:

(a) All the space curves shown in Figures 2 and 3 are continuous because their component functions are continuous at every value of t in $(-\infty, \infty)$.

(b) The function $g(t) = (\cos t)\,\mathbf{i} + (\sin t)\,\mathbf{j} + \lfloor t \rfloor \mathbf{k}$ is discontinuous at every integer, because the greatest integer function $\lfloor t \rfloor$ is discontinuous.

Derivatives and Motion

Suppose that $\mathbf{r}(t) = f(t)\,\mathbf{i} + g(t)\,\mathbf{j} + h(t)\,\mathbf{k}$ is the position vector of a particle moving along a curve in space and that f, g, and h are differentiable functions of t. Then the difference between the particle's positions at time t and time $t + \Delta t$ is the vector $\Delta\mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$,

Definition 4: Derivative and Tangent Vector

Let the point P has position vector $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where f, g and h are differentiable at t. Then $\mathbf{r}$ has a ***derivative*** (or is ***differentiable***) at t and

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{h\to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}.$$

The vector $\mathbf{r}'(t)$ is called the ***tangent vector*** to the curve defined by $\mathbf{r}$ at P, provided $\mathbf{r}'(t)$ exists and $\mathbf{r}'(t) \neq 0$. **The vector $\mathbf{r}'(t)$ points in the direction of the curve at *P*.**

The ***tangent line*** to the curve at P is defined to be the line through P parallel to the tangent vector $\mathbf{r}'(t)$.

Remark:

1. A vector function $\mathbf{r}$ is **differentiable** if it is differentiable at every point of its domain.
2. The curve traced by $\mathbf{r}$ is **smooth** if $\mathbf{r}'(t)$ is continuous and $\mathbf{r}'(t) \neq 0$, that is $f'(t)$, $g'(t)$ and $h'(t)$ are continuous that are not simultaneously zero.

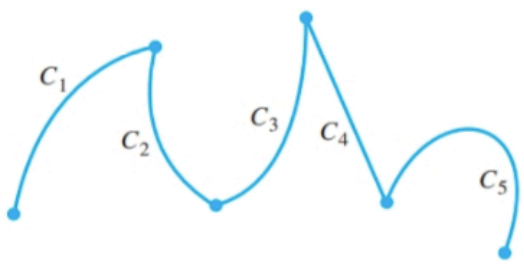


Figure 4 A piecewise smooth curve made up of five smooth curves con-nected end to end in a continuous fashion. The curve here is not smooth at the points joining the five smooth curves.

Example 6:

Find parametric equations for the line tangent to the curve determined by \

$$\mathbf{r}(t) = (e^{-t}\cos t)\mathbf{i} + (e^{-t}\sin t)\mathbf{j} + e^{-t}\mathbf{k} \text{ at time } t = 0.$$

Solution:

$\mathbf{r}(0) = \mathbf{i} + \mathbf{k} \Rightarrow P(1, 0, 1)$ is the point at $t = 0$.

$\mathbf{r}'(t) = (-e^{-t}\cos t - e^{-t}\sin t)\mathbf{i} + (-e^{-t}\sin t + e^{-t}\cos t)\mathbf{j} - e^{-t}\mathbf{k}$

$\Rightarrow \mathbf{r}'(0) = -\mathbf{i} + \mathbf{j} - \mathbf{k}$ is a vector parallel to the tangent line at P.

Parametric equations for the line tangent to the curve at P are

$$x = 1 - t\ ,\ y = t\ ,\ z = 1 - t\ ,\ t \in \mathrm{R}.$$

Velocity, Direction, Speed and Acceleration

Definition 5: If r be the position vector of a particle moving along a smooth curve in space. then

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$$

is the particle's **velocity vector**, tangent to the curve. At any time t, the direction of **v** is the **direction of motion**, the magnitude of **v** is the particle's speed, and the derivative **a** = d**v**/dt, when it exists, is the particle's **acceleration vector**. In summary,

1. Velocity is the derivative of position: $\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$.
2. Speed is the magnitude of velocity: Speed = $|\mathbf{v}|$.
3. Acceleration is the derivative of velocity: $\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$.
4. The unit vector $\frac{\mathbf{v}(t)}{|\mathbf{v}(t)|}$ is the direction of motion at time t.

Example 7: Find the velocity, speed, and acceleration of a particle whose motion in space is given by the position vector $\mathbf{r}(t) = (2\cos t)\,\mathbf{i} + (2\sin t)\mathbf{j} + (5\cos^2 t)\,\mathbf{k}$.

Solution:

The velocity vector is $\mathbf{v}(t) = \mathbf{r}'(t) = (-2\sin t)\,\mathbf{i} + (2\cos t)\mathbf{j} + (-10\cos t \sin t)\mathbf{k}$

$$= (-2\sin t)\,\mathbf{i} + (2\cos t)\mathbf{j} + (-5\sin 2t)\,\mathbf{k}.$$

The acceleration vector is $\mathbf{a}(t) = \mathbf{v}'(t) = (-2\cos t)\,\mathbf{i} + (-2\sin t)\mathbf{j} + (-10\sin 2t)\,\mathbf{k}$.

The speed is $|\mathbf{v}(t)| = \sqrt{(-2\sin t)^2 + (2\cos t)^2 + (-5\sin 2t)^2} = \sqrt{4 + 25\sin^2 2t}$.

Example 8:

Let $\mathbf{r}(t) = (\ln(t^2+1))\mathbf{i} + (\tan^{-1} t)\mathbf{j} + \sqrt{t^2+1}\,\mathbf{k}$. Find the angle between the velocity and acceleration at $t = 0$.

Solution:

The velocity $\mathbf{v}(t) = \mathbf{r}'(t) = \dfrac{2t}{t^2+1}\mathbf{i} + \dfrac{1}{1+t^2}\mathbf{j} + \dfrac{t}{\sqrt{1+t^2}}\mathbf{k}$,

the acceleration $\mathbf{a}(t) = \mathbf{r}''(\mathbf{t}) = \dfrac{2-2t^2}{(t^2+1)^2}\mathbf{i} - \dfrac{2t}{(t^2+1)^2}\mathbf{j} + \dfrac{1}{(t^2+1)^{3/2}}\mathbf{k}$.

$\mathbf{v}(0) = \mathbf{j}$ and $\mathbf{a}(0) = 2\mathbf{i} + \mathbf{k}$.

The angle between $\mathbf{v}(0)$ and $\mathbf{a}(0)$ $= \theta = \cos^{-1}\dfrac{\mathbf{v}(0)\cdot\mathbf{a}(0)}{\|\mathbf{v}(0)\|\|\mathbf{a}(0)\|} = \cos^{-1}\dfrac{0}{1\left(\sqrt{5}\right)} = \dfrac{\pi}{2}$.

Differentiation Rules for Vector Functions

Let $\mathbf{u}$ and $\mathbf{v}$ be differentiable vector functions of t, $\mathbf{C}$ a constant vector, c any scalar, and f any differentiable scalar function.

1. *Constant Function Rule:* $\dfrac{d}{dt}\mathbf{C} = \mathbf{0}$
2. *Scalar Multiple Rules:* $\dfrac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$

   $\dfrac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
3. *Sum Rule:* $\dfrac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$
4. *Difference Rule:* $\dfrac{d}{dt}[\mathbf{u}(t) - \mathbf{v}(t)] = \mathbf{u}'(t) - \mathbf{v}'(t)$
5. *Dot Product Rule:* $\dfrac{d}{dt}[\mathbf{u}(t)\cdot\mathbf{v}(t)] = \mathbf{u}'(t)\cdot\mathbf{v}(t) + \mathbf{u}(t)\cdot\mathbf{v}'(t)$
6. *Cross Product Rule:* $\dfrac{d}{dt}[\mathbf{u}(t)\times\mathbf{v}(t)] = \mathbf{u}'(t)\times\mathbf{v}(t) + \mathbf{u}(t)\times\mathbf{v}'(t)$
7. *Chain Rule:* $\dfrac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

Vector Functions of Constant Length

When a particle moving on a sphere centered at the origin (Figure 5), the position vector has a constant length equal to the radius of the sphere. The velocity vector $d\mathbf{r}/dt$, tangent to the path of motion, is tangent to the sphere and hence perpendicular to $\mathbf{r}$. This is always the case for a differentiable vector function of constant length: The vector and its first derivative are orthogonal. By direct calculation

$$\mathbf{r}(t) \cdot \mathbf{r}(t) = |\mathbf{r}(t)|^2 = c^2$$

$$\Rightarrow \frac{d}{dt}\left[\mathbf{r}(t) \cdot \mathbf{r}(t)\right] = 0$$

$$\Rightarrow \mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$$

$$\Rightarrow 2\mathbf{r}'(t) \cdot \mathbf{r} = 0$$

$$\Rightarrow \mathbf{r}'(t) \cdot \mathbf{r} = 0$$

Thus the vectors $\mathbf{r}'(t)$ and $\mathbf{r}(t)$ are orthogonal because their dot product is 0.

In summary, the following holds.

If $\mathbf{r}$ is a differentiable vector function of t and the length of $\mathbf{r}(t)$ is constant, then

$$\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = 0.$$

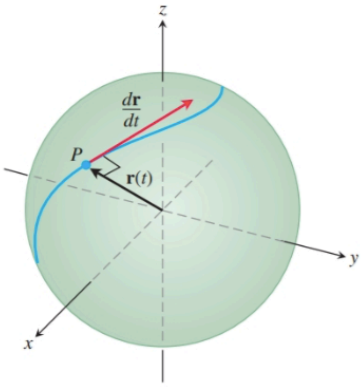


Figure 5 If a particle moves on a sphere in such a way that its position $\mathbf{r}$ is a differentiable function of time, then $\mathbf{r} \cdot (d\mathbf{r}/dt) = 0$.

12.2 Integrals of vector Functions

Definition 6: The indefinite integral of **r** with respect to *t* is the set of all anti-derivatives of **r**, denoted by $\int \mathbf{r}(t)dt$. If **R** is any antiderivative of **r**, then

$$\int \mathbf{r}(t)dt = \mathbf{R}(t) + \mathbf{C} \quad \text{where } \mathbf{C} \text{ is a constant vector.}$$

Remark: Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$. If $F(t)$, $G(t)$ and $H(t)$ are any antiderivatives of $f(t)$, $g(t)$ and $f(t)$, respectively, then

$$\begin{aligned}\int \mathbf{r}(t)dt &= \left(\int f(t)dt\right)\mathbf{i} + \left(\int g(t)dt\right)\mathbf{j} + \left(\int h(t)dt\right)\mathbf{k} \\ &= (F(t) + C_1)\mathbf{i} + (G(t) + C_2)\mathbf{j} + (H(t) + C_3)\mathbf{k} \\ &= (F(t)\mathbf{i} + G(t)\mathbf{j} + H(t)\mathbf{k}) + (C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k} = \mathbf{R}(t) + \mathbf{C}\end{aligned}$$

Example 9:

To integrate a vector function, we integrate each of its components.

$$\begin{aligned}\int\big((\cos t)\mathbf{i} + \mathbf{j} - 2t\mathbf{k}\big)dt &= (\int \cos t\, dt)\mathbf{i} + (\int 1dt)\mathbf{j} - (\int 2tdt)\mathbf{k} \quad \text{---------- (1)} \\ &= (\sin t + C_1)\mathbf{i} + (t + C_2)\mathbf{j} - (t^2 + C_3)\mathbf{k} \quad \text{---------- (2)} \\ &= (\sin t)\mathbf{i} + t\,\mathbf{j} - t^2\mathbf{k} + \mathbf{C}.\end{aligned}$$

As in the integration of scalar functions, we can skip the steps in Equa-tions (1) and (2) and go directly to the final form. Find an antiderivative for each component and add a constant vector at the end.

Definition 7: If the components of $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ are integrable over $[a, b]$, then so is **r**, and the **definite integral** of **r** from *a* to *b* is

$$\int_a^b \mathbf{r}(t)dt = \left(\int_a^b f(t)dt\right)\mathbf{i} + \left(\int_a^b g(t)dt\right)\mathbf{j} + \left(\int_a^b h(t)dt\right)\mathbf{k}.$$

The Fundamental Theorem of Calculus for continuous vector functions says that

$$\int_a^b \mathbf{r}(t)dt = [\mathbf{R}(t)]_a^b = \mathbf{R}(b) - \mathbf{R}(a),$$

where **R** is any antiderivative of **r** so that **R′**(t) = **r**(t). Notice that an antiderivative of a vector function is also a vector function, whereas a definite integral of a vector function is a single constant vector.

Example 10: As in Example 9, we integrate each component.

$$\begin{aligned}\int_0^\pi\big((\cos t)\mathbf{i} + \mathbf{j} - 2t\mathbf{k}\big)dt &= \big(\int_0^\pi \cos t\, dt\big)\mathbf{i} + \big(\int_0^\pi 1dt\big)\mathbf{j} - \big(\int_0^\pi 2tdt\big)\mathbf{k} \\ = [\sin t]_0^\pi\,\mathbf{i} + [t]_0^\pi\,\mathbf{j} - [t^2]_0^\pi\,\mathbf{k} &= (0-0)\mathbf{i} + (\pi - 0)\mathbf{j} - (\pi^2 - 0)\mathbf{k} = \pi\mathbf{j} - \pi^2\mathbf{k}.\end{aligned}$$

Example 11:
Find the position vector $\mathbf{r}(t)$ of a particle if its velocity is $\mathbf{r}(t) = 2t\,\mathbf{i} + (\sin t)\mathbf{j} + (\cos t)\mathbf{k}$ and its initial position vector is $\mathbf{r}(0) = \mathbf{i} + \mathbf{j} + \mathbf{k}$.

Solution:

$$\mathbf{r}(t) = \int \frac{d\mathbf{r}(t)}{dt}dt = \int \mathbf{v}(t)dt$$
$$= (\int 2tdt)\mathbf{i} + (\int \sin t\, dt)\mathbf{j} + (\int \cos t\, dt)\mathbf{k} + \mathbf{C}$$
$$= t^2\,\mathbf{i} - (\cos t)\,\mathbf{j} + (\sin t)\mathbf{k} + \mathbf{C}$$

$\mathbf{r}(0) = \mathbf{i} + \mathbf{j} + \mathbf{k} \Rightarrow 0\mathbf{i} - \mathbf{j} + 0\mathbf{k} + \mathbf{C} = \mathbf{i} + \mathbf{j} + \mathbf{k} \Rightarrow \mathbf{C} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}.$

$\therefore \quad \mathbf{r}(t) = t^2\,\mathbf{i} - (\cos t)\,\mathbf{j} + (\sin t)\mathbf{k} + (\,\mathbf{i} + 2\mathbf{j} + \mathbf{k})$

$= (1 + t^2)\mathbf{i} + (2 - \cos t)\,\mathbf{j} + (1 + \sin t)\mathbf{k}.$

Example 12: Suppose a moving particle departed from the point (4, 0, 0) with velocity $\mathbf{v}(0) = 3\mathbf{j}$. Find the position vector of the particle as a function of t if its acceleration vector is

$$\mathbf{a}(t) = -(3\cos t)\,\mathbf{i} - (3\sin t)\,\mathbf{j} + 2\mathbf{k}.$$

Solution:

The differential equation: $\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = -(3\cos t)\,\mathbf{i} - (3\sin t)\,\mathbf{j} + 2\mathbf{k}$,

The initial conditions: $\mathbf{v}(0) = 3\mathbf{j}$ and $\mathbf{r}(0) = 4\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = 4\mathbf{i}$.

$$\mathbf{v}(t) = \int \frac{d\mathbf{v}(t)}{dt}dt = \int \mathbf{a}(t)dt$$
$$= (\int -3\cos t\, dt)\mathbf{i} + (\int -3\sin t\, dt)\mathbf{j} + (\int 2dt)\mathbf{k} + \mathbf{C}_1$$
$$= -(3\sin t)\,\mathbf{i} + (3\cos t)\,\mathbf{j} + (2t)\mathbf{k} + \mathbf{C}_1$$

$\mathbf{v}(0) = 3\mathbf{j} \Rightarrow 0\mathbf{i} + 3\,\mathbf{j} + 0\mathbf{k} + \mathbf{C}_1 = 3\mathbf{j} \Rightarrow \mathbf{C}_1 = \mathbf{0}.$

$\therefore \quad \mathbf{v}(t) = -(3\sin t)\,\mathbf{i} + (3\cos t)\,\mathbf{j} + (2t)\mathbf{k}.$

$$\mathbf{r}(t) = \int \frac{d\mathbf{r}(t)}{dt}dt = \int \mathbf{v}(t)dt$$
$$= (\int -3\sin t\, dt)\mathbf{i} + (\int 3\cos t\, dt)\mathbf{j} + (\int 2tdt)\mathbf{k} + \mathbf{C}$$
$$= (3\cos t)\,\mathbf{i} + (3\sin t)\,\mathbf{j} + t^2\mathbf{k} + \mathbf{C}$$

$\mathbf{r}(0) = 4\mathbf{i} \Rightarrow 3\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} + \mathbf{C} = 4\mathbf{i} \Rightarrow \mathbf{C} = \mathbf{i}.$

$\therefore$ The position vector of the particle as a function of t is

$$\mathbf{r}(t) = (1 + 3\cos t)\,\mathbf{i} + (3\sin t)\mathbf{j} + t^2\mathbf{k}.$$